Compute Spatiotemporal Moran Eigenvector Maps

Description

Computes Moran Eigenvector Maps (MEMs) from a 3-D coordinate matrix '(x_km, y_km, age_kyr)' built at the **sample level** (one row per site × time-slice). All three dimensions are z-scored before eigenvector computation so that spatial and temporal extents contribute equally to the Euclidean distance structure. Returns the first 'n_mev' eigenvectors as a data frame with row names in '"dataset_name__age"' format, ready for 'scale_spatial_for_fit()' and 'build_jsdm_fit_input()'.

Usage

```
compute_spatiotemporal_mev(

  data_coords_projected = NULL,

  data_sample_ids = NULL,

  n_mev = 20L

)
```

Arguments

`data_coords_projected`	A data frame with 'dataset_name' as row names and columns 'coord_x_km' and 'coord_y_km', as returned by 'project_coords_to_metric()'. One row per unique site/core.
`data_sample_ids`	A data frame with columns 'dataset_name' and 'age' giving the valid (site × time-slice) pairs to model, as returned by 'align_sample_ids()'.
`n_mev`	A positive integer giving the number of eigenvectors to return. If it exceeds the number of positive Moran eigenvectors produced by 'sjSDM::generateSpatialEV()' for the 3-D coordinate matrix, it is automatically clamped down to that count and a 'cli::cli_warn()' message is emitted. Default is '20L'.

Details

Unlike 'compute_spatial_mev()', which operates on unique site coordinates and must be expanded to sample level via 'prepare_spatial_predictors_for_fit()', this function builds the eigenvectors directly on the sample-level 3-D coordinate matrix. This means each observation (site × age) gets a unique row in the MEV matrix, and within-core temporal autocorrelation is captured alongside between-site spatial autocorrelation.

The 3-D coordinate matrix is constructed as follows: 1. Join 'data_coords_projected' (x_km, y_km) onto 'data_sample_ids' to get one row per sample. 2. Convert age (years BP) to kiloyears: 'age_kyr = age / 1000'.

3. Z-score each dimension independently so that the spatial and temporal extents are on the same scale before Euclidean distances are computed.

The z-scoring step is critical: if age is left in kiloyears and coordinates in km, the time axis will dominate or be negligible depending on the data range, producing eigenvectors that are either purely temporal or purely spatial.

Because the eigenvectors are computed at the sample level, **no further expansion step is required**. Pass the returned data frame directly to 'scale_spatial_for_fit()'.

Value

A data frame with row names '"<dataset_name>__<age>"' and 'n_mev' columns named 'mev_1', 'mev_2', …, 'mev_n_mev'. The row order follows 'data_sample_ids' sorted by 'dataset_name' then 'age', matching the ordering produced by 'prepare_abiotic_for_fit()' and 'prepare_community_for_fit()'.

See Also

[compute_spatial_mev()], [project_coords_to_metric()], [align_sample_ids()], [scale_spatial_for_fit()], [build_jsdm_fit_input()]